

KG-COMMIT DESIGN

ONTOLOGY (SCHEMA)

ENTITIES

CORE ENTITIES

TIME

- (DATETIME)

INTERVAL

- ID (String)

COMMIT

- Commit ID (e.g., ac3fb03)

ISSUE

- Issue ID (String)

AUTHOR (DEVELOPER)

- ID (String)

BRANCH

- ID (String)

FILE

- File Name (String)

FILE TYPE

- Format (String): For example, `.py`.

DIR

- Folder Name (String)

EXTERNAL PACKAGE (DEPENDENCY)

- Name (String)

LABEL

- Status: -1 for bug-inducing, 0 for clean, and 1 for bug-fixing.

WITHIN-FILE ENTITIES

CLASS

- Name (String)

FUNCTION

- Name (String)

VARIABLE (DATA)

- Name (String)

DATA TYPE

- ID (String)

FUNCTION SIGNATURE (ARGUMENTS & RETURN)

- ID (String)

FINER ENTITIES (SUB-GRAPHS)

SOURCE CODE

- ???

COMMIT TEXT

- ???

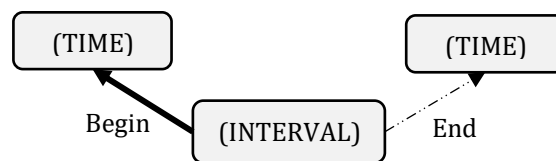
ISSUE TEXT

- ???

RELATIONS

CORE RELATIONS

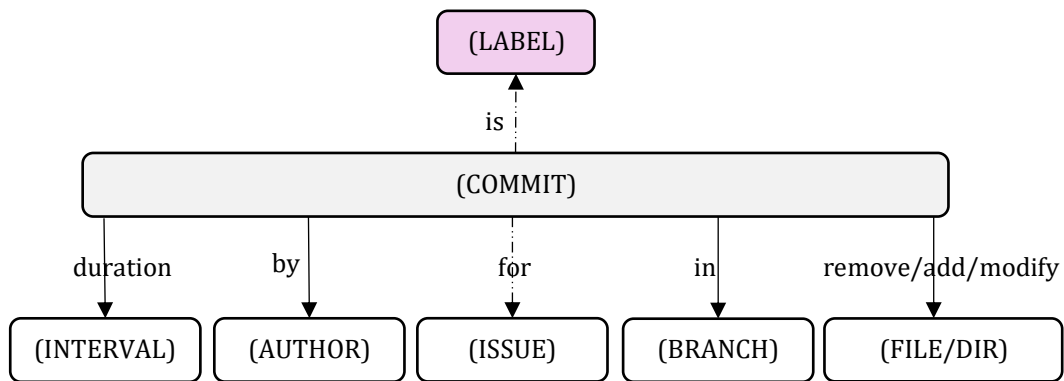
“BEGIN” & “END”: (INTERVAL) → (TIME)



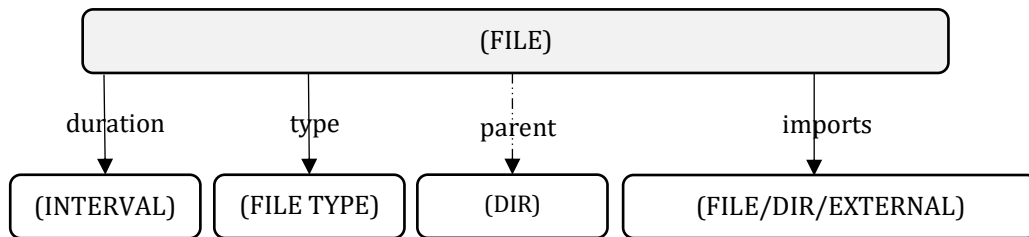
?? → (INTERVAL)

- Possible source nodes: ISSUE, AUTHOR, BRANCH, FILE, DIRECTORY
- Edge name: “duration” or “Within”

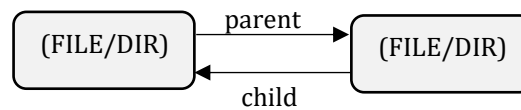
(COMMIT) → ??



(FILE) → ??



FILE AND DIRECTORIES



XXX